BAMEE301 – Computational Engineering

Modules 3 & 4 Assignment – Complete Worked Solutions

All intermediate values are retained to at least four significant digits. Each solution states the governing numerical formulation, the computational steps, an error/convergence check where applicable, and a brief engineering interpretation.

Module 3

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Module 4

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MODULE 3 — Approximation, Differentiation and Integration

1. Newton Divided-Difference Interpolation

> *Q1. Gearbox lubricant viscosity: T(°C)=[25,40,60,85,115], μ(mPa·s)=[162.0,104.8,61.5,31.8,16.4]. Build the full divided-difference table, obtain the 5-point Newton polynomial, estimate μ at 68 °C, and comment on the adequacy of linear interpolation over the full range.*

Divided-difference table

T	$\mu = f[x_i]$	1st order	2nd order	3rd order	4th order
25	162.0	−3.8133	0.047095	−0.0004232	0.00000284
40	104.8	−2.1650	0.021711	−0.0001256	
60	61.5	−1.1880	0.012267		
85	31.8	−0.5133			
115	16.4				

$$P(T) = 162.0 - 3.8133(T-25) + 0.047095(T-25)(T-40) - 0.0004232(T-25)(T-40)(T-60) + 0.00000284(T-25)(T-40)(T-60)(T-85)$$

Evaluation at T = 68 °C

Term order	0	1	2	3	4
Contribution	162.000	−163.973	56.703	−4.075	−0.541

Running sum: 162.000 → −1.973 → 54.730 → 50.655 → **μ(68 °C) ≈ 50.11 mPa·s**

Interpretation: A single linear estimate using only the bracketing points (60,85) gives μ ≈ 51.996 mPa·s, about 1.88 mPa·s (3.7 %) higher than the 5-point value. Because viscosity–temperature behaviour is strongly curved (Arrhenius-type decay), the 2nd/3rd-order divided differences are not negligible — linear interpolation over the full 25–115 °C range is

not adequate; at least a quadratic/cubic fit local to the region of interest should be used for design-accuracy viscosity estimates.

Q2. Thermocouple calibration: E(mV)=[2.10,4.35,6.20,8.75,11.40], T(°C)=[95,188,272,401,548]. Develop the Newton divided-difference polynomial (independent variable E) and estimate T at E = 7.20 mV. Check consistency at a tabulated point.

Divided-difference table

E	$T=f[x_i]$	1st	2nd	3rd	4th
2.10	95.0	41.333	0.99319	0.027779	−0.006629
4.35	188.0	45.405	1.17792	−0.033871	
6.20	272.0	50.588	0.93913		
8.75	401.0	55.472			
11.40	548.0				

$$P(E) = 95.0 + 41.333(E-2.10) + 0.99319(E-2.10)(E-4.35) + 0.027779(E-2.10)(E-4.35)(E-6.20) - 0.006629(E-2.10)(E-4.35)(E-6.20)(E-8.75)$$

Evaluation at E = 7.20 mV

Term contributions: 95.000, 210.800, 14.436, 0.4038, 0.1493 → cumulative sum:

T(7.20 mV) ≈ 320.8 °C

Consistency check

Evaluating the same polynomial at E = 6.20 mV (a tabulated node) reproduces T = 272.0 °C exactly, confirming the divided-difference coefficients and polynomial construction are correct.

Interpretation: The thermocouple response is mildly nonlinear (curvature evident in the 2nd/3rd differences), so the full 4th-order Newton polynomial is preferred over linear calibration for precise temperature recovery from millivolt readings, especially away from the calibration nodes.

2. Lagrange Interpolation

Q3. Brake-pad friction coefficient: μ=[0.42,0.47,0.51,0.46,0.37] at T(°C)=[80,140,200,280,360]. Estimate μ at 225 °C using all 5 points, repeat with the 3 nearest points, and quantify the % difference.

$$L_i(x) = \Pi_{j \neq i} (x-x_j)/(x_i-x_j) \; ; \; P(x) = \Sigma \, y_i L_i(x)$$

Distances of the nodes from x = 225: |225−200|=25, |225−280|=55, |225−140|=85, |225−360|=135, |225−80|=145. The three nearest nodes are T = 140, 200, 280.

Estimate	Nodes used	μ(225 °C)
5-point Lagrange	80,140,200,280,360	0.50759
3-point Lagrange	140,200,280	0.50706

% difference = |0.50759 − 0.50706| / 0.50759 × 100 = **0.104 %**

Interpretation: The two estimates agree to within 0.1 %, showing that near the interpolation region the friction–temperature curve is well approximated locally by a quadratic; the two extra (distant) data points contribute negligibly. A 3-point local Lagrange fit is therefore adequate and computationally cheaper for interpolating within this data set.

Q4. EV battery module: ΔT(°C)=[2.1,4.8,8.9,17.6,30.5] at I(A)=[1.0,2.0,3.0,4.5,6.0]. Estimate ΔT at 3.6 A using (a) a quadratic Lagrange polynomial on the 3 nearest points and (b) a 4th-order polynomial on all 5 points.

Nearest 3 points to I = 3.6 A (by |ΔI|): I = 3.0 (0.6), 4.5 (0.9), 2.0 (1.6).

Estimate	Order	Nodes	ΔT(3.6 A)
Quadratic	2	2.0, 3.0, 4.5	12.013 °C
Full polynomial	4	1.0, 2.0, 3.0, 4.5, 6.0	11.987 °C

Difference = −0.0261 °C (0.22 % of the quadratic estimate) — **ΔT(3.6 A) ≈ 11.99–12.01 °C**

Interpretation: The estimate changes by less than a quarter of one percent between quadratic and quartic order, showing low sensitivity to polynomial order in this region — the temperature-rise curve is smooth (no inflection) near I = 3.6 A, so the cheaper 3-point quadratic is sufficient for battery thermal-management estimates; higher order would matter more near the ends of the range where curvature (I^2R growth) is stronger.

3. Cubic Spline Interpolation

Q5. Elastomeric mount: x(mm)=[0,5,10,15], F(kN)=[0.0,2.8,7.5,15.2]. Construct the natural cubic spline, give the cubic equation on each interval, estimate F at x = 7.5 mm, and verify first-derivative continuity at the internal knots.

Natural-spline system ($M_0=M_3=0$): $h_{i-1}M_{i-1} + 2(h_{i-1}+h_i)M_i + h_iM_{i+1} = 6[(y_{i+1}-y_i)/h_i - (y_i-y_{i-1})/h_{i-1}]$

With h = 5 mm throughout, solving the tridiagonal system gives second derivatives at the knots:

M = [0, 0.07360, 0.16160, 0] kN/mm²

Cubic equations ($S(x) = a + b\cdot\Delta x + c\cdot\Delta x^2 + d\cdot\Delta x^3$, $\Delta x = x - x_i$)

Interval	a	b	c	d
[0,5]	0.0000	0.49867	0.00000	0.002453
[5,10]	2.8000	0.68267	0.03680	0.002933
[10,15]	7.5000	1.27067	0.08080	−0.005387

Estimate at x = 7.5 mm (interval [5,10], Δx = 2.5)

$F(7.5) = 2.8000 + 0.68267(2.5) + 0.03680(2.5)^2 + 0.002933(2.5)^3$ = **4.783 kN**, local slope dF/dx = 0.9217 kN/mm

Derivative continuity check

Knot	Slope from left interval	Slope from right interval
x = 5 mm	0.68267	0.68267
x = 10 mm	1.27067	1.27067

Slopes match exactly to 5 decimals — the spline is C^1 continuous at both interior knots, as required by construction.

Interpretation: The local slope dF/dx is the tangent stiffness of the mount. Stiffness rises from ≈0.499 kN/mm near x=0 to ≈1.27 kN/mm near x=15 mm — a hardening (progressive) elastomer response typical of rubber mounts as internal contact area increases with compression.

Q6. Cam-follower: θ(deg)=[0,15,30,45], s(mm)=[0.0,6.2,14.8,20.0]. Construct the natural cubic spline, estimate displacement and slope at θ = 22.5°, identify the segment used, and explain why a piecewise cubic is preferred over one high-order polynomial.

M = [0, 0.023111, −0.028444, 0] mm/deg²

Interval	a	b	c	d
[0,15]	0.0000	0.35556	0.00000	0.000257
[15,30]	6.2000	0.52889	0.01156	−0.000573
[30,45]	14.8000	0.48889	−0.01422	0.000316

θ = 22.5° lies in the segment [15°, 30°] (Δθ = 7.5°):

s(22.5°) = 6.2000 + 0.52889(7.5) + 0.01156(7.5)² − 0.000573(7.5)³ = **10.575 mm**, slope ds/dθ = 0.6056 mm/deg (≈34.7 mm/rad)

Interpretation: A single 3rd-degree polynomial can be forced through 4 points, but as more cam-profile stations are added, one global high-degree polynomial develops spurious oscillations between data points (Runge-type behaviour) and a local perturbation anywhere on the cam affects the fit everywhere. The piecewise cubic spline instead gives C^1/C^2 smooth, locally controlled segments — essential for a cam profile where jerk/acceleration continuity governs follower dynamics.

4. Linear Regression and Curve Fitting

Q7. Tensile test: ε(×10⁻³)=[0.5,1.0,1.5,2.0,2.5,3.0], σ(MPa)=[102,201,304,397,506,598]. Fit σ = a + bε by least squares, obtain E, R², and interpret the intercept.

b = [nΣxy − ΣxΣy] / [nΣx² − (Σx)²] ; a = (Σy − bΣx)/n ; $R^2 = 1 - SS_{res}/SS_{tot}$

With x = ε (actual strain, dimensionless): n=6, Σx=0.010500, Σy=2108, Σxy=4.561, Σx²=2.275×10⁻⁵.

b = 199,314 MPa (Young's modulus E ≈ **199.3 GPa**, consistent with steel)
a = 2.533 MPa R² = **0.99967**

Interpretation: R² ≈ 0.9997 confirms excellent linear (Hookean) behaviour in this stress range. The small positive intercept (2.53 MPa, ideally zero) reflects minor instrument zero-offset/pre-load rather than a physical phenomenon, since stress must vanish at zero strain.

Q8. Coast-down test: ω(rad/s)=[60,90,120,150,180,210], T(N·m)=[8.4,10.1,12.5,14.0,16.2,17.8]. Fit T = a + bω, estimate T at 145 rad/s, find R², and interpret a, b physically.

a = 4.578 N·m b = 0.063619 N·m·s/rad R² = **0.99707**
T(145 rad/s) = 4.578 + 0.063619(145) = **13.80 N·m**

Interpretation: Intercept a = 4.578 N·m represents the speed-independent Coulomb (dry-friction) resisting torque; slope b = 0.0636 N·m·s/rad is the viscous-damping coefficient, i.e. the torque increase per unit angular speed from lubricant/

windage drag. $R^2 = 0.997$ confirms the coast-down torque is dominated by these two mechanisms with little unmodeled nonlinearity.

5. Polynomial Regression

Q9. Progressive spring: x(mm)=[0,10,20,30,40,50], F(kN)=[0.2,2.0,5.1,9.7,16.1,24.6]. Fit $F = a_0+a_1x+a_2x^2$ by least squares, estimate F at x = 32 mm, find R^2, and compare tangent vs average stiffness over 20–40 mm.

$a_0 = 0.3321$ kN, $a_1 = 0.065607$ kN/mm, $a_2 = 0.0083393$ kN/mm² $R^2 =$ **0.99969**

$F(32\text{ mm}) = 0.3321 + 0.065607(32) + 0.0083393(32)^2 =$ **10.97 kN**

Quantity	Definition	Value
Tangent stiffness at x=32	$dF/dx = a_1+2a_2x$	0.5993 kN/mm
F(20 mm)	—	4.980 kN
F(40 mm)	—	16.299 kN
Average (secant) stiffness 20–40 mm	[F(40)−F(20)]/20	0.5660 kN/mm

Interpretation: The tangent stiffness at x = 32 mm (0.599 kN/mm) exceeds the average secant stiffness over 20–40 mm (0.566 kN/mm) by ≈ 5.9 %, confirming a convex, hardening (progressive) load–deflection curve — physically typical of a variable-pitch/conical progressive spring where stiffness increases with compression.

Q10. Wind-tunnel drag: V(m/s)=[10,15,20,25,30,35], F(N)=[42,91,158,248,357,489]. Fit linear and quadratic models, compare RSS/R^2, estimate drag at 27 m/s, and justify the better model mechanically.

Model	Coefficients	R^2	SS_{res}	F(27 m/s)
Linear: $F=a+bV$	a=−170.70, b=17.846	0.97176	4048.4	311.1 N
Quadratic: $F=a_0+a_1V+a_2V^2$	a_0=9.757, a_1=−0.8936, a_2=0.41643	0.99999	2.121	289.2 N

Quadratic fit reduces residual sum of squares by ~1900× and raises R^2 from 0.9718 to 0.99999 — **the quadratic model is decisively superior**.

Interpretation: Aerodynamic drag follows $F_d = \frac{1}{2}\rho C_d A V^2$, i.e. is physically quadratic in speed; the near-perfect quadratic R^2 confirms this drag law, while the linear fit's poor tail behaviour (large residuals at high V) is a direct symptom of forcing a straight line onto a V^2 relationship. The quadratic estimate (289.2 N at 27 m/s) is the mechanically justified answer.

6. Numerical Differentiation — Forward, Backward, Central Differences

Q11. Beam: x(m)=[0,0.25,0.50,0.75,1.00], y(mm)=[0.000,0.920,2.180,3.960,6.420]. Estimate slope at x=0 (2nd-order forward), at x=1.0 (2nd-order backward), at x=0.5 (central); find curvature at x=0.5 and, for E=200 GPa, $I=6.0\times10^{-6}$ m^4, estimate bending moment.

Forward: $y'_0 = (-3y_0+4y_1-y_2)/(2h)$ Backward: $y'_n = (3y_n-4y_{n-1}+y_{n-2})/(2h)$ Central: $y'_i = (y_{i+1}-y_{i-1})/(2h)$

h = 0.25 m

Location	Formula used	Slope (mm/m)

x = 0	(−3(0)+4(0.920)−2.180)/0.5	3.000
x = 1.00	(3(6.420)−4(3.960)+2.180)/0.5	11.20
x = 0.50	(3.960−0.920)/0.5	6.080

Curvature at x = 0.5 m

$y'' = (y_{i+1} - 2y_i + y_{i-1})/h^2 = (3.960 - 2(2.180) + 0.920)/0.25^2 = 8.320$ mm/m² (= 0.008320 m⁻¹ after converting y to metres)

Bending moment

$M = EI \cdot y'' = (200\times10^9)(6.0\times10^{-6})(0.008320) =$ **≈ 9.98 kN·m**

Interpretation: The slope grows from 3.0×10^{-3} rad at the fixed end to 11.2×10^{-3} rad at the free end, consistent with a cantilever-type deflection profile; the curvature-derived bending moment (≈10 kN·m at midspan) can be cross-checked against the applied loading using beam theory $M=EI\ d^2y/dx^2$.

Q12. Machine foundation, Δt=0.01 s: t=[0…0.06], x(mm)=[0.000,0.412,0.775,1.041,1.176,1.158,0.982]. Estimate velocity at t=0.03 s by forward, backward and central differences, and acceleration by central difference. Compare and discuss noise amplification.

Method	Formula (indices about t=0.03s)	Velocity (mm/s)
Forward (2nd order)	$(-3x_3+4x_4-x_5)/(2\Delta t)$	21.15
Backward (2nd order)	$(3x_3-4x_2+x_1)/(2\Delta t)$	21.75
Central	$(x_4-x_2)/(2\Delta t)$	20.05

Central-difference acceleration: $a = (x_4-2x_3+x_2)/\Delta t^2 =$ **−1310 mm/s²**

Interpretation: The three velocity estimates differ by up to ≈8 % because the follower is near its displacement peak (t≈0.04 s) where the curvature is highest, so one-sided differences carry larger truncation error than the central estimate. Second-derivative (acceleration) estimates amplify measurement noise far more than first derivatives, since dividing by Δt^2 (here 10^{-4} s²) magnifies any small error in x by ~10^4 — at Δt=0.01 s this data would need to be reasonably clean/smoothed before trusting the acceleration value for foundation design.

7. Numerical Integration — Trapezoidal Rule

Q13. Crash-energy absorber: x(mm)=[0,5,…,50] (11 pts), F(kN)=[0.0,1.1,2.6,4.8,7.9,12.0,17.2,23.6,31.5,41.0,52.3]. Estimate absorbed energy by composite trapezoidal rule; repeat with every alternate point; discuss the effect of step size.

Composite trapezoidal: $I \approx h[\tfrac{1}{2}y_0 + y_1+\ldots+y_{n-1} + \tfrac{1}{2}y_n]$

Grid	h (mm)	Points used	Energy (kN·mm = J)
All data	5	11	839.25
Alternate points	10	6	853.50

% change = (853.50−839.25)/839.25 × 100 = **+1.70 %**

Interpretation: The F–x curve is convex (increasing slope); trapezoidal segments always lie above a convex curve, so coarsening the grid (h: 5→10 mm) increases the over-estimate. The 1.7 % change confirms the finer 5 mm grid is markedly more accurate and should be preferred for reporting absorbed energy.

Q14. Drag force $F(x)=900e^{0.35x}+120\sin(2x)$ N over $0\le x\le 2$ m. Use composite trapezoidal rule with n=4 and n=8; compare as a discretization-sensitivity measure.

n	h (m)	Estimated work (J)
4	0.50	2704.25
8	0.25	2705.60

Difference = 1.35 J (0.050 % of the n=8 value) — **estimate has essentially converged**

Interpretation: Trapezoidal error is $O(h^2)$; halving h reduced the estimated discretization error by roughly a factor of four (consistent theory), and the very small 0.05 % change indicates n=8 is already sufficiently fine for this smooth exponential-plus-sinusoid integrand.

8. Numerical Integration — Simpson's Rule

Q15. Using the Q13 crash-absorber data, estimate absorbed energy by Simpson's 1/3 rule; compare with the Q13 trapezoidal estimate at the same spacing and explain why the two need not agree for nonlinear experimental data.

Composite Simpson 1/3 (n=10 even intervals, h=5): $I \approx (h/3)[y_0+y_n+4\Sigma y_{odd}+2\Sigma y_{even,interior}]$

Simpson estimate = **834.50 J** (Trapezoidal at h=5 mm: 839.25 J)

% difference = $|839.25-834.50|/834.50 \times 100$ = **0.569 %**

Interpretation: Simpson's rule assumes the curve is locally quadratic over each pair of intervals, whereas the trapezoidal rule assumes local linearity; because the measured F–x response is not an exact quadratic (real experimental convex curve with varying local curvature), the two methods integrate slightly different fitted shapes through the same points and need not coincide exactly — here they agree to within 0.6 %, and Simpson's estimate is generally the more accurate of the two for smooth, convex data.

Q16. Rotary damper torque $T(\theta)=45+18\sin\theta+12\sin(3\theta)+0.8\theta^2$ N·m, $0\le\theta\le\pi$. Estimate work by Simpson's 1/3 rule with n=6 and n=12; compare and comment on convergence.

n	h (rad)	Work (J)
6	π/6	194.033
12	π/12	193.659

Change = −0.374 J (0.19 %) — **monotonic convergence, consistent with Simpson's $O(h^4)$ error**

Interpretation: Doubling the number of intervals reduced the change by roughly the expected 16× ($O(h^4)$) factor relative to a coarser refinement step, confirming the numerical result is converging rapidly toward the true work value; n=12 can be treated as sufficiently accurate for damper energy-dissipation estimates.

9. Gaussian Quadrature

Q17. Actuator force $F(x)=1800e^{-0.6x}\sin(2.5x)+700$ N over $0\le x\le 1.2$ m. Estimate work using 1-, 2- and 3-point Gauss–Legendre quadrature; compare and explain why finite-point Gauss quadrature is not exact for this integrand.

$\int_a^b f(x)dx \approx [(b-a)/2]\ \Sigma\ w_i\ f(x_m+[(b-a)/2]\xi_i)$, mapping $[a,b]\rightarrow[-1,1]$

Points	Nodes ξ_i (weights w_i)	Work estimate (J)
1	0 (2)	2343.21
2	±0.5774 (1,1)	1817.98
3	0, ±0.7746 (8/9, 5/9, 5/9)	1837.78

Successive change: 1-pt → 2-pt = −525.2 J (22.4 %); 2-pt → 3-pt = +19.8 J (1.09 %) — **rapid convergence toward ≈1838 J**

Interpretation: An n-point Gauss rule is exact only for polynomials up to degree (2n−1). Because F(x) contains a product of an exponential and a sine (a transcendental, non-polynomial function), no finite-point Gauss rule can integrate it exactly — each additional point only improves the polynomial approximation of F(x), so the estimates converge toward, but never exactly equal, the true work.

Q18. $I = \int_0^2 (3+2x+x^2+0.5x^4)\,dx$. Evaluate using 1-, 2-, 3-point Gauss–Legendre quadrature after mapping to $[-1,1]$; compare with the analytical integral and relate to the $2n-1$ exactness property.

Points	Estimate	Exact (analytical)
1	13.000	238/15 = 15.867
2	15.778	
3	15.867	

3-point Gauss quadrature reproduces the analytical value 238/15 = 15.867 exactly

Interpretation: The integrand is a degree-4 polynomial. A 3-point Gauss rule is exact for polynomials up to degree $2(3)-1 = 5 \ge 4$, so it integrates this integrand exactly, matching the analytical result to machine precision — a direct demonstration of the 2n−1 exactness property. The 1-point and 2-point rules (exact only to degree 1 and 3 respectively) cannot capture the quartic term and show visible error.

MODULE 4 — PDE Classification and Initial-Value Problems

10. Classification of PDEs — Parabolic, Elliptic, Hyperbolic

Q19. Classify, using the discriminant B^2-4AC on $Au_{xx}+Bu_{xy}+Cu_{yy}=0$, and give a mechanical-engineering application for: (i) $T_{xx}+4T_{xy}+T_{yy}=0$, (ii) $4u_{xx}+4u_{xy}+u_{yy}=0$, (iii) $u_{xx}-4u_{yy}=0$.

Equation	A, B, C	$\Delta=B^2-4AC$	Classification	Application
(i) $T_{xx}+4T_{xy}+T_{yy}=0$	1, 4, 1	$16-4=12>0$	**Hyperbolic**	Transient stress-wave / vibration propagation in an anisotropic medium
(ii) $4u_{xx}+4u_{xy}+u_{yy}=0$	4, 4, 1	$16-16=0$	**Parabolic**	Directional/anisotropic diffusion, e.g. transient heat conduction along a preferred axis
(iii) $u_{xx}-4u_{yy}=0$	1, 0, −4	$0-4(1)(-4)=16>0$	**Hyperbolic**	Vibrating membrane / string with unequal wave speeds along x and y

Interpretation: A positive discriminant (hyperbolic) signals two real characteristic directions along which information/ disturbances propagate at finite speed (wave-like behaviour); $\Delta=0$ (parabolic) signals a single characteristic direction typical of diffusion/smoothing processes; a negative discriminant (elliptic, not occurring above) would signal equilibrium/ boundary-value behaviour with no real characteristics.

Q20. Classify (i) $\nabla^2T=0$, (ii) $T_t=\alpha T_{xx}$, (iii) $u_{tt}=c^2u_{xx}$. State the initial/boundary data normally required and distinguish equilibrium, diffusion and wave propagation.

Equation	Standard form / A,B,C	Δ	Class	Data required
(i) $T_{xx}+T_{yy}=0$	A=1,B=0,C=1	$0-4=-4<0$	**Elliptic**	Boundary conditions on the entire closed boundary (Dirichlet/Neumann); no time variable — steady state
(ii) $T_t-\alpha T_{xx}=0$	A=α (x-term), 1st order in t	degenerate (single family)	**Parabolic**	One initial condition T(x,0) + boundary conditions at both x-ends for all t>0
(iii) $u_{tt}-c^2u_{xx}=0$	$A=c^2$,B=0,C=−1	$0-4c^2(-1)=4c^2>0$	**Hyperbolic**	Two initial conditions, u(x,0) and u_t(x,0), plus boundary conditions at both ends

Interpretation: **Elliptic** problems (steady 2-D conduction) describe equilibrium states governed entirely by boundary data, solved simultaneously over the whole domain. **Parabolic** problems (transient 1-D conduction) describe diffusion: disturbances spread instantaneously (infinite propagation speed) but smooth out with time, solved by marching forward from an initial state. **Hyperbolic** problems (wave propagation) describe finite-speed transport of disturbances along characteristic lines, needing both a displacement and a velocity initial condition, and preserving sharp fronts rather than smoothing them.

11. Initial-Value Problems — Euler's Method

Q21. Flywheel: J=3 kg·m², T(t)=24+8sin(2t) N·m, viscous resistance 5ω, ω(0)=1 rad/s. Formulate the IVP; solve with explicit Euler, h=0.1 s to t=0.5 s; repeat with h=0.05 s; compare final values.

$J\,d\omega/dt = T(t) - 5\omega \implies d\omega/dt = [24+8\sin(2t)-5\omega]/3,\ \omega(0)=1$ Euler: $\omega_{i+1}=\omega_i+h\cdot f(t_i,\omega_i)$

t (s)	0	0.1	0.2	0.3	0.4	0.5
ω, h=0.1 (rad/s)	1.0000	1.6333	2.2141	2.7489	3.2413	3.6924

With h = 0.05 s (10 steps): **ω(0.5) = 3.6536 rad/s** vs h=0.1: ω(0.5) = 3.6924 rad/s

Interpretation: Halving the step size changes ω(0.5) by 0.0388 rad/s (≈1.05 %), consistent with Euler's first-order O(h) global error roughly halving when h is halved. The flywheel speed rises steadily toward a quasi-steady oscillation about the mean torque/damping balance ($\bar{T}/5\omega \Rightarrow \omega \approx 4.8$ rad/s at steady state).

Q22. Casing thermal model: $dT/dt = -0.08(T-25)+3e^{-0.5t}$, T(0)=90 °C. Euler with h=0.5 min to t=3 min; repeat h=0.25 min; compare T(3); explain each ODE term physically.

t (min)	0	0.5	1.0	1.5	2.0	2.5	3.0
T, h=0.5 (°C)	90.00	88.90	87.51	85.92	84.19	82.38	80.51

h=0.25 min (12 steps): **T(3) = 80.33 °C** vs h=0.5 min: T(3) = 80.51 °C (difference 0.19 °C, 0.23 %)

Interpretation: The term −0.08(T−25) is Newtonian cooling of the casing toward the 25 °C ambient with lumped thermal conductance 0.08 min^{-1}; the term $+3e^{-0.5t}$ is a decaying internal heat-generation source (e.g. residual friction/windage after shutdown) that fades with a ~2-minute time constant. Halving h reduces the estimate by only 0.23 %, showing the Euler solution has essentially converged and h=0.5 min is adequate for this slowly-varying process.

12. Runge–Kutta Methods

Q23. Damped actuator: $dy/dt = -1.5y+2\sin(3t)$, y(0)=0.8. Use classical RK4, h=0.1 s, to t=0.4 s. Tabulate k_1–k_4 at every step and compare with Euler (same h).

$k_1=f(t,y)$; $k_2=f(t+h/2,y+hk_1/2)$; $k_3=f(t+h/2,y+hk_2/2)$; $k_4=f(t+h,y+hk_3)$; $y_{i+1}=y_i+(h/6)(k_1+2k_2+2k_3+k_4)$

Step (t → t+h)	k_1	k_2	k_3	k_4	y at t+h
0.0 → 0.1	−1.2000	−0.8111	−0.8403	−0.4829	0.71690
0.1 → 0.2	−0.4843	−0.1691	−0.1927	0.0828	0.69815
0.2 → 0.3	0.0821	0.3099	0.2928	0.4755	0.72753
0.3 → 0.4	0.4754	0.6079	0.5980	0.6831	0.78704

RK4: y(0.4) = 0.78704

t	0	0.1	0.2	0.3	0.4
Euler y	0.8000	0.6800	0.6371	0.6545	0.7130

Interpretation: At t=0.4 s, RK4 (0.7870) and Euler (0.7130) differ by 0.0741 (≈10 %) at this coarse h=0.1 s — Euler's O(h) local truncation error accumulates rapidly for this oscillatory-damped system, whereas RK4's O(h^5) local error makes it dramatically more accurate at the same step size. RK4 should be preferred whenever accuracy at moderate step size is required.

Q24. Rotor: $d\omega/dt = 12-0.8\omega-0.04\omega^2$, ω(0)=2 rad/s. Use RK2 (midpoint) and RK4, h=0.2 s, to t=1.0 s. Compare and discuss the effect of method order for this nonlinear IVP.

t (s)	0	0.2	0.4	0.6	0.8	1.0

ω, RK2 (rad/s)	2.0000	3.8430	5.3198	6.4767	7.3672	8.0436
ω, RK4 (rad/s)	2.0000	3.8541	5.3411	6.5054	7.4002	8.0779

ω(1.0): RK2 = 8.0436, RK4 = 8.0779 rad/s **difference = 0.0342 rad/s (0.42 %)**

Interpretation: The two methods agree closely because the nonlinear ω² damping term varies smoothly over this range; nonetheless, RK4's higher order (local error $O(h^5)$ vs RK2's $O(h^3)$) makes it systematically more accurate, and that advantage would widen for larger h or a more strongly nonlinear right-hand side (e.g. near saturation of ω). For routine engineering estimates at this h, RK2 is already within ~0.4 % of RK4.

13. Adams–Bashforth Predictor Method

Q25. $dy/dt = -2y + \cos t$. Starting values $y(0.0)=1.000000$, $y(0.1)=0.900000$, $y(0.2)=0.819500$, $y(0.3)=0.755019$. Use the 4-step Adams–Bashforth predictor to estimate $y(0.4)$ and $y(0.5)$. Explain why a multistep method needs starting values.

AB4: $y_{i+1} = y_i + (h/24)(55f_i - 59f_{i-1} + 37f_{i-2} - 9f_{i-3})$

t	**0.0**	**0.1**	**0.2**	**0.3**
y	1.000000	0.900000	0.819500	0.755019
f=−2y+cos t	−1.00000	−0.80500	−0.65893	−0.55470

$y(0.4) = 0.755019 + (0.1/24)[55(-0.55470) - 59(-0.65893) + 37(-0.80500) - 9(-1.00000)]$ = **0.70328**

Using $f(0.4) = -2(0.70328) + \cos(0.4) = -0.48957$, advance again:

$y(0.5) = 0.70328 + (0.1/24)[55(-0.48957) - 59(-0.55470) + 37(-0.65893) - 9(-0.80500)]$ = **0.65699**

Interpretation: A 4-step Adams–Bashforth formula needs f at the four most recent nodes (weights 55,−59,37,−9), so it cannot self-start from a single initial condition — three additional starting values (here at t=0.1, 0.2, 0.3) must first be generated by a self-starting single-step method such as RK4. The AB4 predictions (0.7033, 0.6570) closely track the exact analytical solution $y = 0.2\sin t + 0.4\cos t + 0.6e^{-2t}$ (0.7159, 0.6676), confirming the method's $O(h^4)$ accuracy.

Q26. Rotor speed $d\omega/dt = 8 + 2\sin t - 0.6\omega$, $h=0.1$ s. Given $\omega(0.0)=2.0000$, $\omega(0.1)=2.6650$, $\omega(0.2)=3.2921$, $\omega(0.3)=3.8865$, use 4-step AB to predict $\omega(0.4)$ and $\omega(0.5)$; comment on how starting-value error propagates.

t	**0.0**	**0.1**	**0.2**	**0.3**
ω	2.0000	2.6650	3.2921	3.8865
f=8+2sin t−0.6ω	6.8000	6.6007	6.4221	6.2591

$\omega(0.4) = 3.8865 + (0.1/24)[55(6.2591) - 59(6.4221) + 37(6.6007) - 9(6.8000)]$ = **4.5047**

$f(0.4) = 8 + 2\sin(0.4) - 0.6(4.5047) = 6.1084$; advancing again:

$\omega(0.5) = 4.5047 + (0.1/24)[55(6.1084) - 59(6.2591) + 37(6.4221) - 9(6.6007)]$ = **5.1010**

Interpretation: Because AB4 forms a weighted combination of the four most recent derivative values, any error present in the given starting values is carried directly into every subsequent prediction. Here the ODE is stable (the −0.6ω term damps deviations), so a small starting error tends to decay rather than grow; for a less stable (or stiff/unstable) system, the

same starting error could instead amplify over successive steps, so accurate self-starting values (e.g. from RK4) are essential for reliable multistep results.

14. Adams Predictor–Corrector Method

Q27. $dy/dt = -y + 0.5\sin(2t)$, $h = 0.1$ s. Starting values $y(0.0) = 1.000000$, $y(0.1) = 0.909675$, $y(0.2) = 0.836857$, $y(0.3) = 0.779050$. Predict $y(0.4)$ with 4-step AB, correct with Adams–Moulton, perform a second correction, and quantify the change between successive corrected values.

AM corrector: $y_{i+1} = y_i + (h/24)(9f_{i+1} + 19f_i - 5f_{i-1} + f_{i-2})$

t	0.0	0.1	0.2	0.3
y	1.000000	0.909675	0.836857	0.779050
f=−y+0.5sin(2t)	−1.00000	−0.81034	−0.64215	−0.49673

AB4 predictor: $y^P(0.4) = \mathbf{0.735650}$

$f^P(0.4) = -0.735650 + 0.5\sin(0.8) = -0.377336$. Apply AM corrector:

1st correction: $y^{C1}(0.4) = \mathbf{0.735591}$

Re-evaluating f with y^{C1} and correcting once more:

2nd correction: $y^{C2}(0.4) = \mathbf{0.735593}$

Change between successive corrected values: $|y^{C2} - y^{C1}| = \mathbf{2.23 \times 10^{-6}}$ — negligible

Interpretation: The predictor and first corrected value already differ by only ~6×10^{-5}, and the second correction changes the result by just 2.2×10^{-6} — well below any practical engineering tolerance. This shows the predictor–corrector pair has essentially converged after a single correction for this smooth, well-behaved (non-stiff) ODE.

Q28. Thermal response $dT/dt = -0.12(T - 30) + 4\cos(0.5t)$, $h = 0.5$ min. Given $T(0) = 85.000$, $T(0.5) = 83.120$, $T(1.0) = 81.392$, $T(1.5) = 79.818$ °C, use Adams-Bashforth predictor and Adams-Moulton corrector to estimate $T(2.0$ min$)$, correcting until successive values differ by less than 0.001 °C; report the number of corrections needed.

t	0.0	0.5	1.0	1.5
T	85.000	83.120	81.392	79.818
f=−0.12(T−30)+4cos(0.5t)	−2.6000	−2.4988	−2.6567	−3.0514

AB4 predictor: $T^P(2.0) = \mathbf{78.1485}$ °C

Iteration	Corrected T(2.0)	|ΔT| (°C)
Correction 1	78.15672	0.008203
Correction 2	78.15654	0.000185 (< 0.001 ✓ converged)

T(2.0 min) ≈ 78.157 °C, reached after 2 corrector iterations

Interpretation: The predictor–corrector loop converges in only two passes, reflecting the mild, non-stiff nature of the lumped cooling model. Physically, the casing continues cooling toward the 30 °C ambient (Newtonian term) while the oscillatory 4cos(0.5t) term supplies a slowly-varying internal heat input; by t=2 min the predicted temperature (78.16 °C) shows the cooling trend beginning to be gently offset by the periodic heating term.

End of solutions — BAMEE301 Modules 3 & 4 Assignment. All values computed to at least 4 significant figures using the governing numerical formulations shown.